

Session 2018-2019.

1. A.

- I. what are the advantages and disadvantages of digital techniques over analog technique.

Advantages of Digital Techniques over Analog Techniques:

1. **Storage:** Digital data can be stored in a variety of formats and devices, such as hard drives, flash drives, or cloud storage, making it easier to store and retrieve large amounts of data.
2. **Signal Processing:** Digital signals can be easily manipulated and processed using software, making it easier to filter, amplify, or modulate them.
3. **Reproduction:** Digital signals can be reproduced with no loss of quality, unlike analog signals, which degrade over time due to noise, distortion, or interference.
4. **Flexibility:** Digital signals can be easily converted into different formats or standards, making it possible to use the same data across multiple devices or platforms.

Disadvantages of Digital Techniques over Analog Techniques:

1. **Complexity:** Digital systems are generally more complex and require more processing power, storage capacity, and software to operate efficiently.
2. **Cost:** Digital systems can be more expensive than analog systems due to the need for specialized hardware and software.
3. **Sampling Rate:** Digital signals require a higher sampling rate to accurately capture the analog signal, which can lead to larger file sizes and longer processing times.
4. **Non-linear Distortion:** Digital signals can suffer from non-linear distortion, which occurs when the signal is amplified beyond its maximum level, leading to clipping or distortion.

II. Simplify the Boolean function

$$f(w,x,y,z) = \sum(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

wx/yz	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

All cells are completely filled by value hence the output is 1.

B.

- I. Prove that $f_1 = m_0 + m_2 + m_5 + m_7 = M_1 \cdot M_3 \cdot M_4 \cdot M_6$.
- II. Find the complement of the function $F_1 = x'yz' + x'y'z$ and $F_2 = x(y'z' + yz)$ applying De Morgan's theorem.

$$F_1 = x'yz' + x'y'z$$

$$\begin{aligned} F_1' &= (x'yz' + x'y'z)' \\ &= (x'yz')'(x'y'z)' \\ &= (x+y+z)(x+y+z') \end{aligned}$$

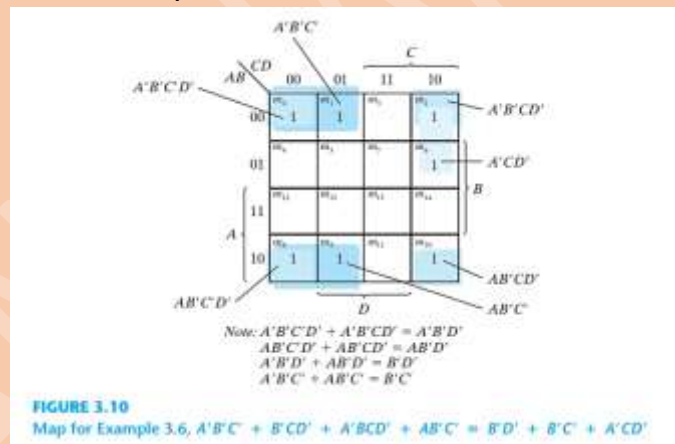
$$F_2 = x(y'z' + yz)$$

$$\begin{aligned} F_2' &= (xy'z' + xyz)' \\ &= (xy'z')'(xyz)' \\ &= (x' + y + z)(x' + y' + z') \end{aligned}$$

2.

- A. simplify the Boolean function $F = A'B'C' + B'CD' + A'BCD' + AB'C'$ and implement the resultant using logic gates.

➤ For simplify we use k-map and find:



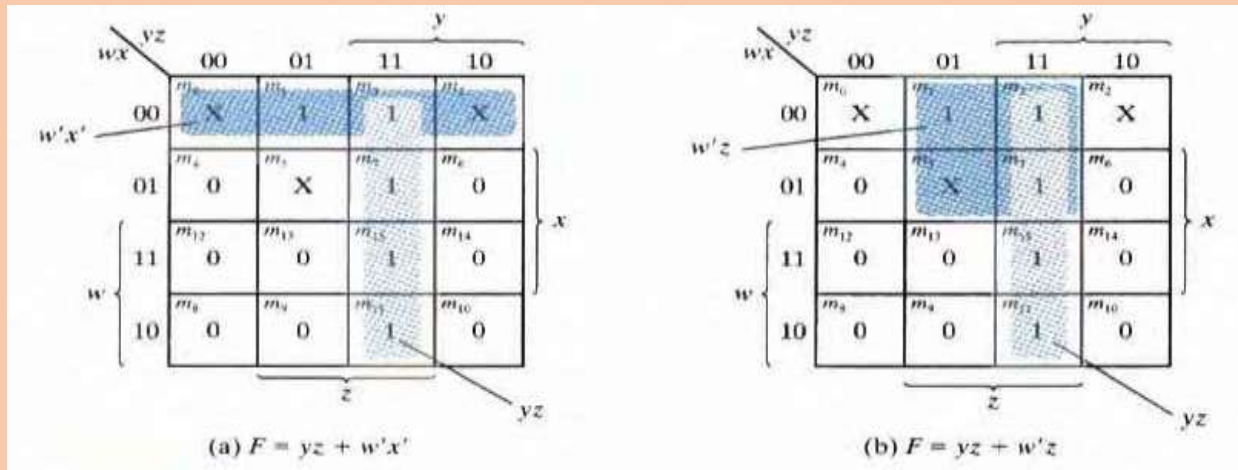
Simplified form is, $F = B'D' + B'C' + A'CD'$

➤ Now we can easily draw the circuit diagram for this simplified expression.

- B. "An expression with the minimum number of literals is not necessarily unique"- Illustrate the significance of the above statement with appropriate example.

Consider an example-Simplify the Boolean function $F(w, x, y, z) =$

$\Sigma(1, 3, 7, 11, 15)$ which has the don't-care conditions $d(w, x, y, z) = \Sigma(0, 2, 5)$.



- ✓ In part (a) of the diagram, don't-care minterms 0 and 2 are included with the 1's, resulting in the simplified function: **$F = yz + w'x'$**
- ✓ In part (b) don't-care minterm 5 is included with the 1's and the simplified function is now **$F = yz + w'z$** .
 - An expression with the minimum number of literals is not necessarily unique.

3.

A. A four bit binary number is represented as $A_3A_2A_1A_0$, where A_3, A_2, A_1 and A_0 represent the individual bits and A_0 is equal to the LSB. Design a logic circuit that will produce a HIGH output whenever the binary number is greater than 0010 and less than 1000.

Solⁿ:

- The Truth Table for the required operation is as shown:

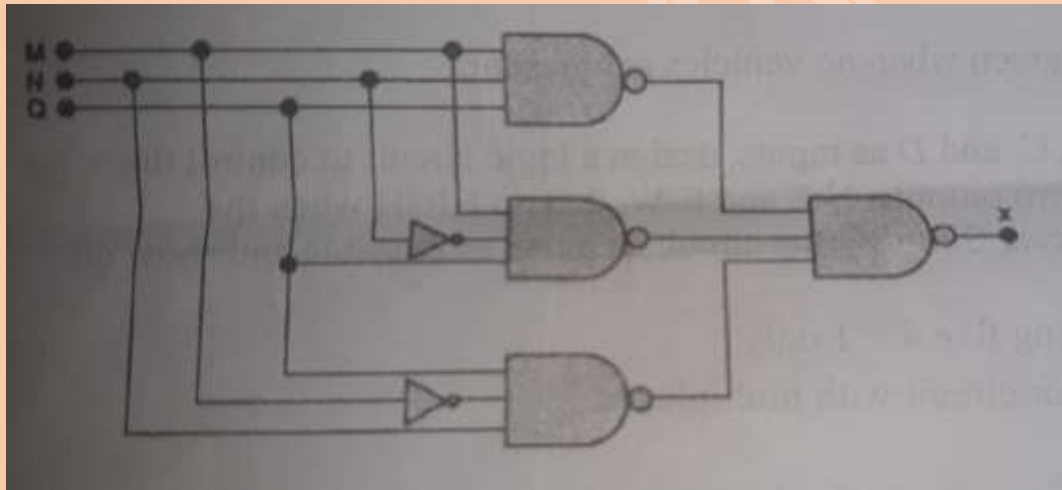
A_3	A_2	A_1	A_0	F(Output)
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

- Using K-map to simplify the expression, we get:

		$A_1 A_0$			
		00	01	11	10
$A_3 A_2$	00	0	1	1	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

$$F = \bar{A}_3 A_2 + \bar{A}_3 A_1 A_0$$

B.



- I. Simplify the circuit of figure using Boolean algebra.

From the above circuit we find:

$$\begin{aligned}
 F &= [(MNQ)'(MN'Q)'(M'NQ)']' \\
 &= (MNQ)'' + (MN'Q)'' + (M'NQ)' \\
 &= MNQ + MN'Q + M'NQ \\
 &= MQ + M'NQ \\
 &= Q(M + M'N) \\
 &= Q(M + N) \\
 &= MQ + QN
 \end{aligned}$$

- II. Change each gate in problem to a NOR gate and simplify the circuit using Boolean algebra.

Now we can implement the simplified expression using NOR gates.

4.

A. Design a circuit that produces a HIGH output only when all three inputs are the same level.

I. Use a truth table and k-map to produce the SOP solution.

Input A	Input B	Input C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

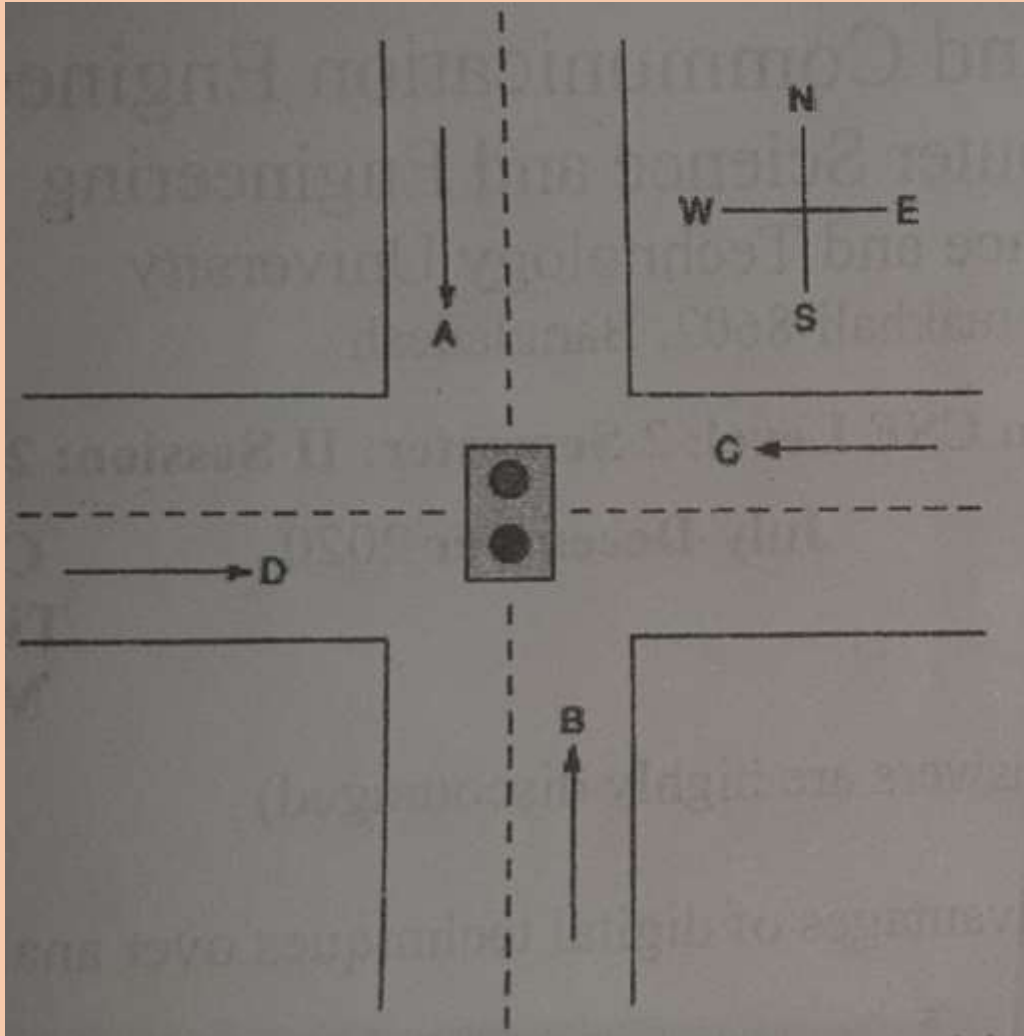
II. Use two-input XOR and other gates to find a solution.

Now, We can draw the logic diagram using two input XOR and other gates.

B. Four large tanks at a chemical plant contain different liquids being heated. Liquid-level sensors are being used to detect whenever the level in tank A and tank B rises above a predetermined level. Temperature sensors in tanks C and D detect when the temperature in either of these tanks drops below a prescribed temperature limit. Assume that the liquid-level is too high. Also, the temperature sensor outputs C and D are LOW when the temperature is satisfactory and HIGH when the temperature is too low. Design a logic circuit that will detect whenever the level in tank A or tank B is too high at the same time that the temperature in either tank C or tank D is too low.

✓ https://www.youtube.com/watch?v=CdrqChUQ-YA&ab_channel=IsaIshaq

5.



- A. The above figure shows the intersection of a main highway with a secondary access road. Vehicle-detection sensors are placed along lanes C and D (main road) and lanes A and B (access road). These sensor outputs are LOW(0) when no vehicle is present and HIGH(1) when a vehicle is present. The intersection traffic light is to be controlled according to the following logic:
- I. The East-west (E-W) traffic light will be green whenever both lanes C and D are occupied.
 - II. The E-W light will be green whenever either C or D is occupied but lanes A and B are not both occupied.
 - III. The North-south (N-S) light will be green whenever both lanes A and B are occupied but C and D are not both occupied.
 - IV. The N-S light will also be green when either A or B is occupied while C and D are both vacant.
 - V. The E-W light will be green when no vehicles are present.

Using the sensor outputs A,B,C and D as inputs, design a logic circuit to control the traffic light. There should be two outputs, N-S and E-W , that go HIGH the corresponding light is to be green. Simplify the circuit as much as possible and show all steps.

➤ **Truth table:**

Inputs				Outputs	
A	B	C	D	N-S	E-W
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	0	1
0	1	1	0	0	1
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	0	1
1	0	1	0	0	1
1	0	1	1	0	1
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	0	1

3. **MAP:**

AB \ CD	00	01	11	10
00	X	X		X
01	1			
11	1	1		1
10	1			

1	1	1	1	0	1
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$$N - S = BC'D' + AC'D' + AB C' + AB D'$$

or

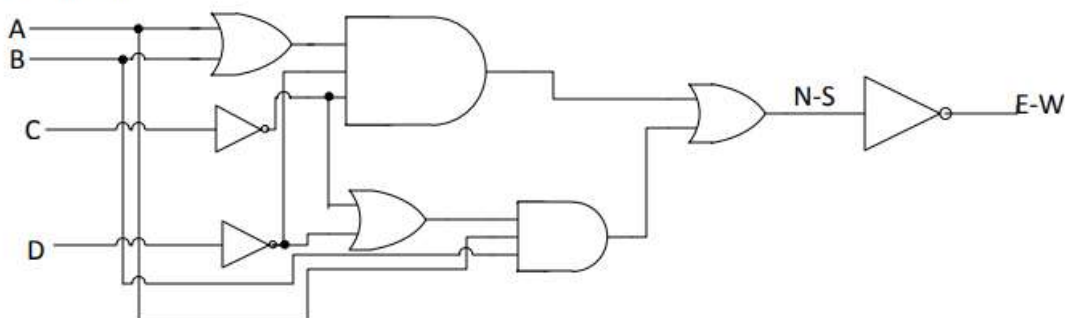
$$N - S = C'D'(A + B) + AB(C' + D')$$

Look at the truth table, The output of E-W is just opposite of the output of N-S. So we can write the following equation for E-W:

$$E - W = (N - S)'$$

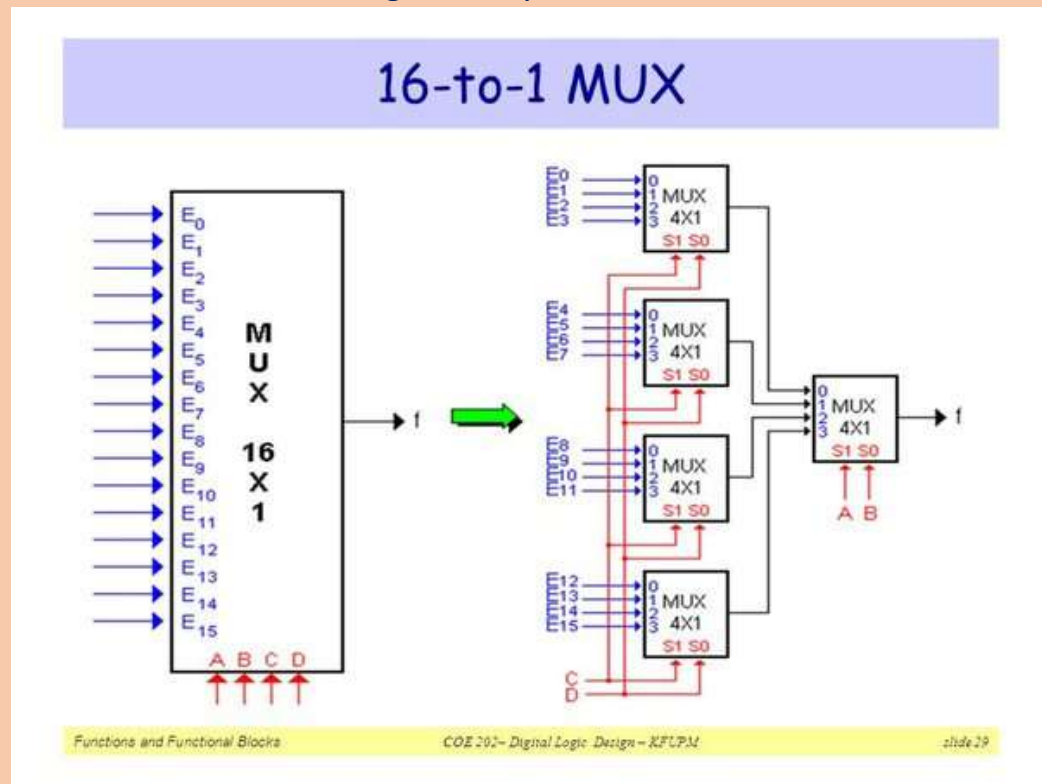
Fig: Map for N-S

4. **Circuit Diagram :**



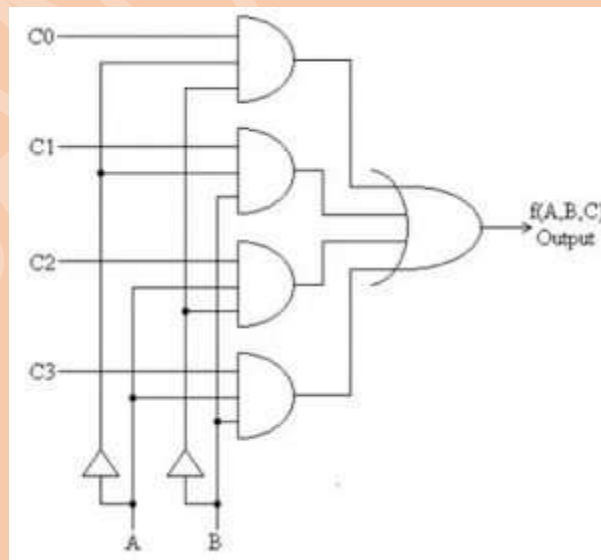
B.

I. Construct 16x1 mux using 4x1 only.



II. Implement full-subtractor circuit with multiplexer.

The design of this using 4X1 multiplexer is shown in the following logic diagram. This design can be done using the following steps.



- In step1, there are two outputs like Sub and Borrow. So we have to choose 2 multiplexers.
- In step2, the truth table can be implemented along with K-maps

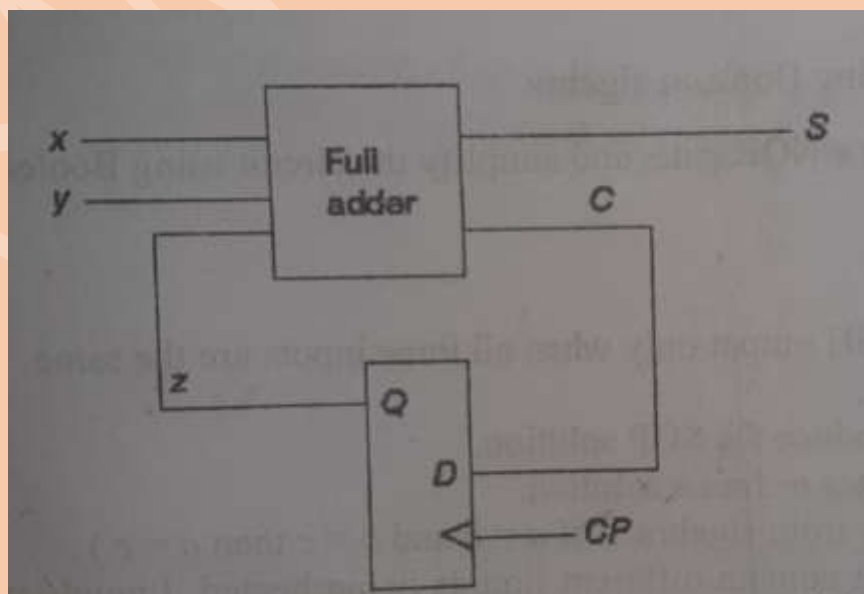
- In step3, the two variables can be selected as your select line. For instance, B & C are in this case.

Truth Table

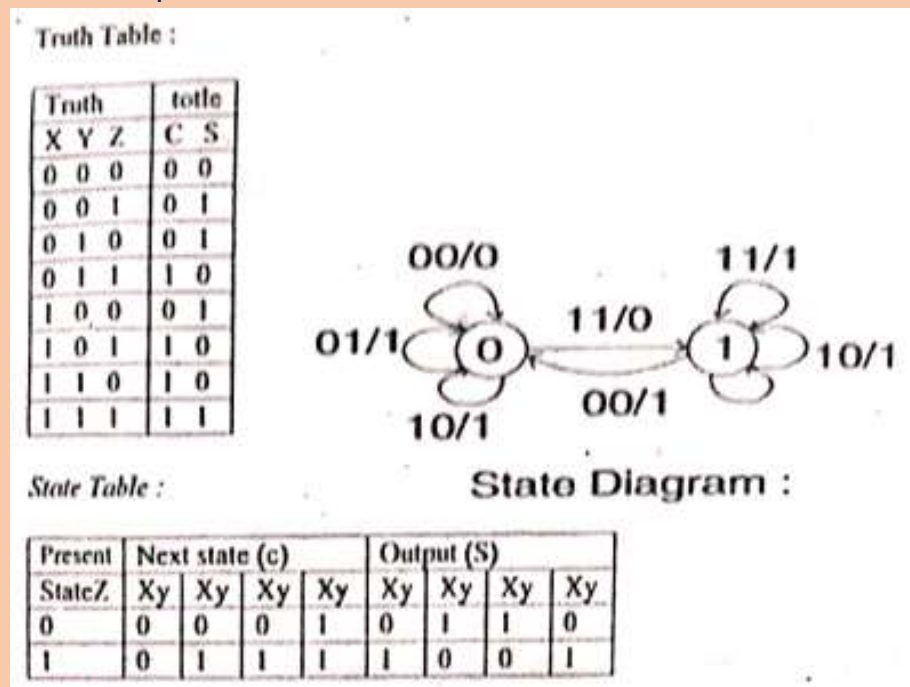
The truth table of full subtractor circuit using 4X1 multiplexer includes the following

A	B	C	Sub	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

6.



- A. The full adder of figure receives two external inputs x and y ; the third input z comes from the output of a D flip-flop. The carry output is transferred to the flip-flop every clock pulse. The external S output gives the sum of x , y and z . Obtain the state table and state diagram of the sequential.



- B. Consider a JK' flip-flop, i.e, a JK flip-flop with an inverter between external input K' and internal input K .

I. Obtain the flip-flop characteristic table.

Q	J	K'	$Q(t+1) = JQ' + K'Q$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

II. Obtain the characteristic equation.

We can derive the equation from above equation.

III. Show that tying the two external inputs together forms of a D flip-flop.

C. Define shift register. Design a serial transfer circuit with an example.

A shift register is a type of digital circuit using a cascade of flip-flops where the output of one flip-flop is connected to the input of the next.

